

Layer Normalization in Transformers

Deep Learning

Daniel E. Acuna

Associate Professor, University of Colorado Boulder

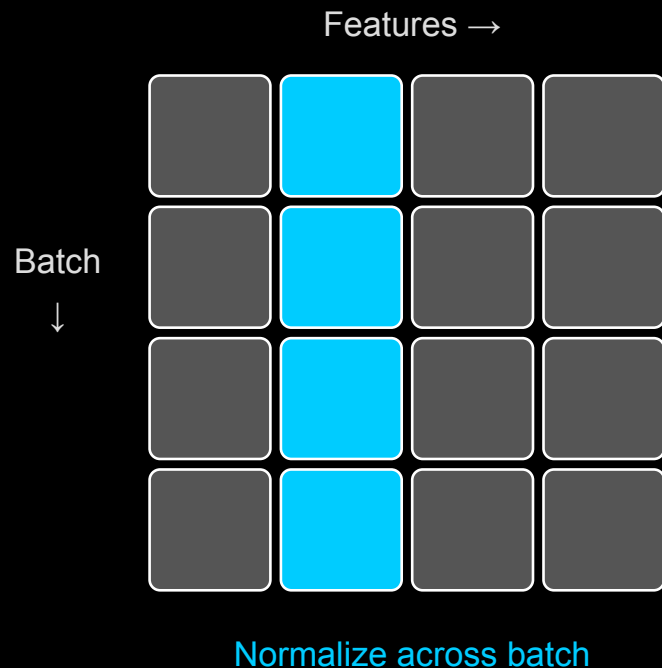
Contents of This Video

In this video, we will cover:

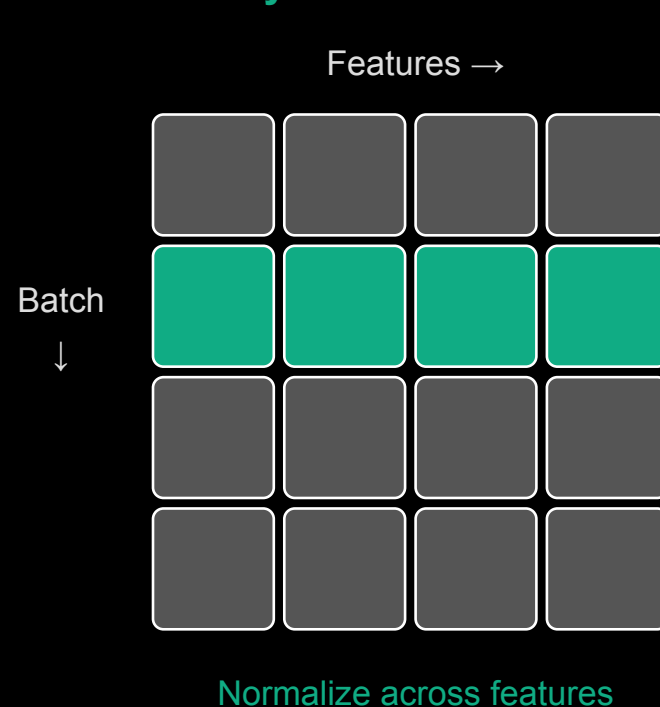
- Batch norm vs. layer norm
- Why batch norm doesn't work for sequences
- Layer normalization formula
- Pre-norm vs. post-norm Transformers
- Practical benefits of layer norm

Batch Norm vs. Layer Norm

Batch Normalization



Layer Normalization



Why Batch Norm Fails for Sequences

Problem 1: Variable sequence lengths

- Batch norm needs consistent positions across batch
- Sequences have different lengths
- Padding positions would contaminate statistics

Problem 2: Batch size dependency

- Batch norm requires large batches
- Long sequences → small batches (memory)
- Statistics become unreliable

Layer Normalization Formula

$$\text{LayerNorm}(x) = \gamma \odot \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta$$

Where (computed per sample, across features):

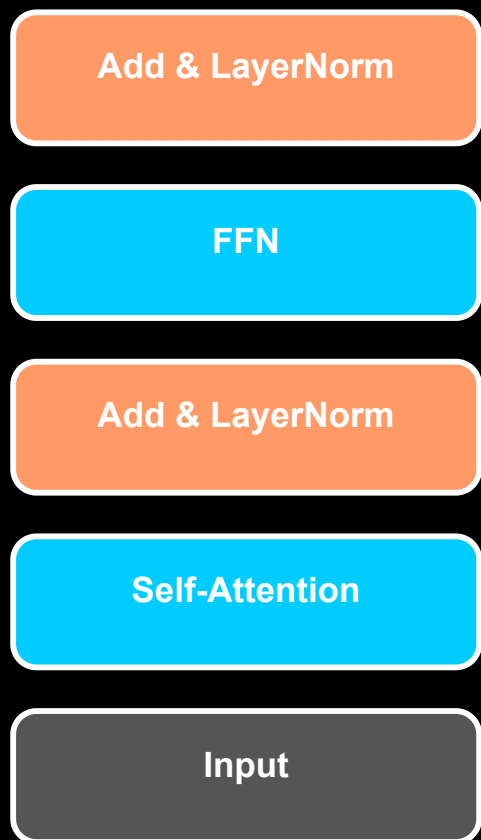
$$\mu = \frac{1}{d} \sum_{i=1}^d x_i$$

$$\sigma^2 = \frac{1}{d} \sum_{i=1}^d (x_i - \mu)^2$$

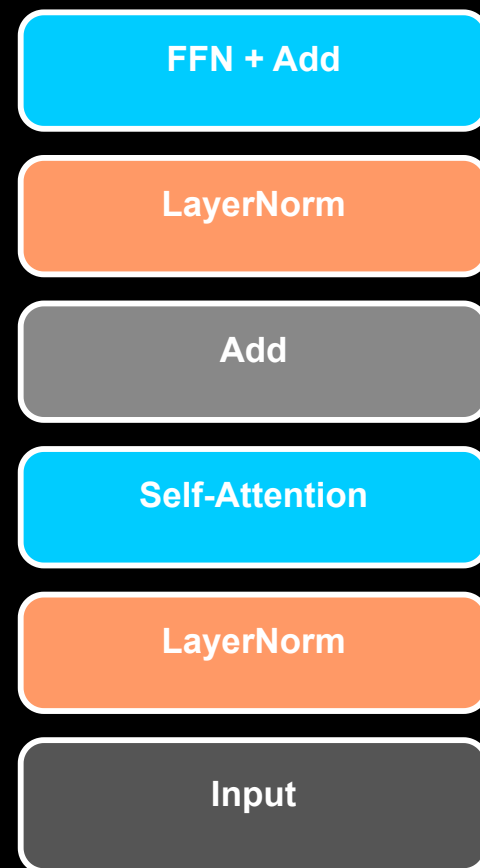
γ and β are learnable parameters

Layer Norm in Transformers

Post-LN (Original Transformer)



Pre-LN (GPT, Modern)



Benefits of Layer Normalization

Stabilizes training:

- Keeps activations in reasonable range
- Prevents exploding/vanishing activations
- Enables higher learning rates

Works with any sequence length:

- No dependency on batch statistics
- Each position normalized independently
- Works at inference with batch size 1

Standard in all Transformers

What We've Covered

In this video, we've learned:

- Batch norm: normalize across batch (bad for sequences)
- Layer norm: normalize across features (good for sequences)
- Formula: $\gamma \odot \frac{x - \mu}{\sigma} + \beta$ (computed per sample)
- Post-LN (original) vs Pre-LN (modern, more stable)
- Layer norm stabilizes training, works with any sequence length